

Project FORESIGHT

AI-Powered Demand & Inventory Intelligence Platform

Notebook 03: Feature Engineering

Project Information

Item	Details
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1. Business Context and Notebook Objective

Demand forecasting requires more than clean historical data. Forecasting models need structured predictors that represent seasonality, recent demand behaviour, longer-term demand patterns, pricing, promotions and other information available before the forecast period.

Notebook 03 transforms the validated daily analytical base produced by Notebook 02 into a weekly, leakage-safe modelling dataset for the approved 300 SKU-store series.

The objectives of this notebook are to:

- create one complete weekly observation for each SKU-store series;
- define weekly units sold as the forecasting target;
- engineer historical demand, calendar, price, promotion and demand-behaviour features;
- identify weekly observations with sufficient preceding history for model eligibility;
- validate feature calculations and prevent future-information leakage;
- export the final dataset and feature metadata for forecasting.

This notebook performs feature engineering only. Forecasting models, validation periods, performance metrics and inventory recommendations will be developed in subsequent notebooks.



2. Notebook Workflow and Feature Engineering Framework

Notebook 03 follows a focused feature-engineering workflow designed to transform the validated daily analytical base produced by Notebook 02 into a weekly, leakage-safe modelling dataset for demand forecasting.

2.1 Notebook Workflow

Section	Focus
3	Notebook Setup and Input Data
4	Weekly Modelling Base
5	Forecasting Feature Engineering
6	Final Modelling Dataset
7	Feature Validation and Leakage Audit
8	Export and Validate Modelling Outputs
9	Conclusion and Notebook 04 Handoff

2.2 Feature Engineering Framework

The modelling dataset will be constructed around the following components:

Component	Purpose
Weekly Target	Define demand as weekly units sold for each SKU-store series
Calendar Features	Represent time progression and recurring seasonal patterns using forecast-time-available calendar information
Historical Demand Features	Represent prior demand behaviour through lagged and rolling statistics across short-, medium- and annual horizons
Commercial and Calendar Drivers	Represent historical price and promotion conditions together with scheduled-event and applicable SNAP information known before the target week
Demand-Behaviour Features	Describe prior-demand recency, activity and intermittency at each target week
Hierarchy Features	Preserve item, category, department, store and state context for later modelling
History Eligibility	Identify series-week observations with sufficient preceding history for model use

2.3 Feature Engineering Principles

- Use only the validated 300 SKU-store development scope.
- Build one complete weekly observation for each SKU-store series.
- Use weekly units sold as the forecasting target.
- Derive demand and commercial features only from information available before the target week.
- Use directly only those calendar variables that are known in advance.
- Treat missing values created by lags and rolling windows as structural.
- Retain categorical hierarchy fields for later encoding within the modelling pipeline.
- Keep simulated inventory variables outside the core demand-forecasting dataset.
- Create only features with a clear forecasting purpose.

Scope boundary: This notebook prepares and validates forecasting features only. Evaluation periods, forecasting models, performance metrics and inventory decisions belong to subsequent notebooks.

3. Notebook Setup and Input Data

This section establishes the reproducible notebook environment, defines the feature-engineering configuration and loads the validated daily analytical base produced by Notebook 02. Formal input-contract and structural-readiness checks will verify that the dataset is suitable for weekly feature engineering.

3.1 Configure the Notebook

This subsection establishes the required Python environment, project paths, modelling input contract and central feature-engineering parameters. An input-availability check will confirm that the validated Notebook 02 handoff is accessible before it is loaded.

3.1.1 Library Imports and Display Settings

The required libraries will support data manipulation, numerical feature construction, project-path management and metadata export. Notebook-wide display settings will also be configured to keep validation outputs readable.

✅ Required libraries imported and display settings configured.

Library Summary

The required Python libraries were successfully imported and configured for Notebook 03. Each library supports a specific part of the feature-engineering workflow.

Library	Purpose
pandas	Data loading, weekly aggregation, feature construction and validation
NumPy	Numerical calculations, cyclical transformations and missing-value handling
pathlib	Project-directory and file-path management
json	Export of approved feature lists and supporting metadata

The notebook environment is now prepared for input validation, weekly dataset construction and leakage-safe feature engineering.

3.1.2 Project Paths and Input Contract

Notebook 03 will use the validated daily analytical base exported by Notebook 02 as its single modelling input. Centralised project paths will support reproducible input loading and later output generation.

The modelling-output directory is defined here but will not be created until the completed outputs are exported in Section 8.

```
Project root           : C:\Users\hp\Project FORESIGHT
Interim data directory : C:\Users\hp\Project FORESIGHT\data\interim
Modeling output directory : C:\Users\hp\Project FORESIGHT\data\modeling
Analytical base        : C:\Users\hp\Project FORESIGHT\data\interim\analytics_base_daily.parquet
```

3.1.3 Feature-Engineering Configuration

The approved development scope, minimum history requirement and historical-demand windows will be defined centrally. These parameters will be reused consistently during feature construction and validation.

The feature groups and governing principles are fixed, but the final feature list will be confirmed only after the available history and individual feature behaviour have been examined.

	parameter	value
0	Expected SKU-store series	300
1	Minimum history requirement	52 weeks
2	Demand lags	1, 2, 4, 8, 13, 26, 52
3	Rolling-mean windows	4, 8, 13, 26, 52
4	Rolling-standard-deviation windows	4, 13, 52

3.1.4 Input Availability Check

The required Notebook 02 handoff will be checked before it is loaded. This prevents feature engineering from beginning with a missing or incorrectly located input file.

	dataset	relative_path	exists	file_size_mb
0	Daily analytical base	data\interim\analytics_base_daily.parquet	True	4.60

✅ Required Notebook 02 handoff is available.

✓ Configuration Summary

Notebook 03 is now configured for reproducible feature engineering:

- the required libraries and display settings have been established;
- the project root and modelling paths have been defined;
- the validated daily analytical base has been established as the single modelling input;
- the development scope and historical-demand parameters have been centralised;
- the required Notebook 02 handoff has been located successfully.

The notebook environment is ready to load and verify the daily analytical base.

3.2 Load and Verify the Daily Analytical Base

The validated daily analytical base produced by Notebook 02 will now be loaded as the single input to the feature-engineering workflow.

Before any transformation is performed, the dataset will be checked against its documented handoff contract. The validation will cover:

- dataset dimensions and date coverage;
- required modelling fields;
- data types and missingness;
- uniqueness and completeness of the daily analytical key;
- preservation of the approved 300-series development scope;
- balanced daily panel coverage;
- chronological ordering and target validity.

These checks create a controlled boundary between the validated analytical layer and the downstream modelling layer.

3.2.1 Load the Validated Analytical Base

The Parquet handoff will be loaded without modifying its values or structure. A separate working dataset will be created later when weekly transformations begin.

✓ Daily analytical base loaded successfully.

Rows : 573,900

Columns : 56

3.2.2 Input Snapshot

A concise snapshot will establish the loaded dataset's dimensions, memory footprint, series coverage and date coverage before detailed validation begins.

	metric	value
0	Rows	573,900
1	Columns	56
2	In-memory size (MB)	313.18
3	Unique SKU-store series	300
4	Unique dates	1,913
5	Minimum date	2011-01-29 00:00:00
6	Maximum date	2016-04-24 00:00:00

3.2.3 Schema and Missingness Inspection

The complete input schema will be inspected before modelling fields are selected. This check records each column's data type, non-null coverage and missing-value count without altering the dataset.

	column	dtype	non_null_rows	missing_rows	missing_pct
0	date	datetime64[us]	573900	0	0.00
1	sku_id	str	573900	0	0.00
2	units_sold	int64	573900	0	0.00
3	revenue	float64	573900	0	0.00
4	unit_price	float64	573900	0	0.00
5	promo_flag	int64	573900	0	0.00
6	d	str	573900	0	0.00
7	wm_yr_wk	int64	573900	0	0.00
8	year	int64	573900	0	0.00
9	quarter	int64	573900	0	0.00
10	month	int64	573900	0	0.00
11	month_name	str	573900	0	0.00
12	week_of_year	int64	573900	0	0.00
13	day	int64	573900	0	0.00
14	day_name	str	573900	0	0.00
15	day_of_week	int64	573900	0	0.00
16	is_weekend	int64	573900	0	0.00
17	season	str	573900	0	0.00
18	event_name_1	str	46200	527700	91.95
19	event_type_1	str	46200	527700	91.95
20	event_name_2	str	1200	572700	99.79
21	event_type_2	str	1200	572700	99.79
22	event_flag	int64	573900	0	0.00
23	snap_CA	int64	573900	0	0.00
24	snap_TX	int64	573900	0	0.00
25	snap_WI	int64	573900	0	0.00
26	snap_flag	int64	573900	0	0.00
27	item_id	str	573900	0	0.00
28	category	str	573900	0	0.00
29	department	str	573900	0	0.00
30	store_id	str	573900	0	0.00
31	state_id	str	573900	0	0.00
32	launch_date	str	573900	0	0.00
33	first_sale_date	str	573900	0	0.00
34	last_sale_date	str	573900	0	0.00
35	sku_age_days	int64	573900	0	0.00
36	days_since_last_sale	int64	573900	0	0.00
37	avg_price	float64	573900	0	0.00
38	max_observed_price	float64	573900	0	0.00
39	estimated_unit_cost	float64	573900	0	0.00
40	estimated_gross_margin	float64	573900	0	0.00
41	sku_status	str	573900	0	0.00
42	opening_stock	int64	573900	0	0.00
43	fulfilled_units	int64	573900	0	0.00
44	lost_sales_units	int64	573900	0	0.00
45	received_stock	int64	573900	0	0.00
46	order_placed_units	int64	573900	0	0.00
47	closing_stock	int64	573900	0	0.00
48	reorder_point	int64	573900	0	0.00
49	reorder_quantity	int64	573900	0	0.00
50	on_order_units	int64	573900	0	0.00
51	lead_time_days	int64	573900	0	0.00
52	target_stock	int64	573900	0	0.00
53	inventory_status	str	573900	0	0.00
54	estimated_inventory_value	float64	573900	0	0.00
55	days_of_supply	float64	573900	0	0.00

	dtype	column_count
0	int64	28
1	str	19
2	float64	8
3	datetime64[us]	1

3.2.4 Core Input Contract

The analytical base is expected to preserve the verified Notebook 02 handoff and provide the key, chronology and target fields required for preliminary structural validation.

This preliminary contract is intentionally narrower than the complete weekly source-field contract enforced in Section 3.2.7.

The contract distinguishes between:

- **snapshot expectations**, which reconcile the loaded file with the documented Notebook 02 export;
- **core structural requirements**, which must hold before detailed panel validation can proceed.

	contract_item	expected_value	observed_value
0	Expected input rows	573,900	573,900
1	Expected input columns	56	56
2	Expected daily dates	1,913	1,913
3	Expected date range	2011-01-29 to 2016-04-24	2011-01-29 00:00:00 to 2016-04-24 00:00:00
4	Expected SKU-store series	300	300
5	Required modelling fields	sku_id, date, units_sold	All available

3.2.5 Structural and Panel Validation

The following checks will verify that the input remains a complete and uniquely keyed daily panel. These conditions are required before observations can be aggregated into weekly modelling records.

✅ Core modelling fields are available.

Date dtype: datetime64[us]

	metric	observed_value
0	Duplicate sku_id-date keys	0
1	Missing analytical-key values	0
2	Unique SKU-store series	300
3	Unique dates	1913
4	Minimum records per series	1913
5	Maximum records per series	1913
6	Minimum series per date	300
7	Maximum series per date	300
8	Missing target values	0
9	Negative target values	0

3.2.6 Chronology Inspection

A chronological sample will confirm that the analytical key, target and selected source attributes align correctly at the daily grain. The complete dataset will remain unchanged; sorting will be applied only to the displayed sample.

	sku_id	date	units_sold	unit_price	promo_flag	event_name_1	event_flag	snap_flag
0	FOODS_1_001_CA_1	2011-01-29	3	2.00	0	NaN	0	0
1	FOODS_1_001_CA_1	2011-01-30	0	2.00	0	NaN	0	0
2	FOODS_1_001_CA_1	2011-01-31	0	2.00	0	NaN	0	0
3	FOODS_1_001_CA_1	2011-02-01	1	2.00	0	NaN	0	1
4	FOODS_1_001_CA_1	2011-02-02	4	2.00	0	NaN	0	1
5	FOODS_1_001_CA_1	2011-02-03	2	2.00	0	NaN	0	1
6	FOODS_1_001_CA_1	2011-02-04	0	2.00	0	NaN	0	1
7	FOODS_1_001_CA_1	2011-02-05	2	2.00	0	NaN	0	1
8	FOODS_1_001_CA_1	2011-02-06	0	2.00	0	SuperBowl	1	1
9	FOODS_1_001_CA_1	2011-02-07	0	2.00	0	NaN	0	1

3.2.7 Input Validation Register

The executed input checks will now be consolidated into a formal validation register. Each control compares the observed analytical base against either the documented Notebook 02 snapshot or a structural requirement of the weekly feature-engineering pipeline.

The required source-field contract includes only the variables needed to construct the weekly target, hierarchy, commercial, event and SNAP features. Inventory-derived fields and current-period revenue are not approved as forecasting predictors at this stage.

	control	expected	observed	status
0	Snapshot row count	573,900	573,900	PASS
1	Snapshot column count	56	56	PASS
2	Daily date count	1,913	1,913	PASS
3	Date range	2011-01-29 to 2016-04-24	2011-01-29 to 2016-04-24	PASS
4	Required weekly source fields	All 20 fields available	All available	PASS
5	Date data type	Datetime-compatible	datetime64[us]	PASS
6	Unique sku_id-date key	0 duplicates	0 duplicates	PASS
7	Analytical-key completeness	0 missing values	0 missing values	PASS
8	SKU-store development scope	300 series	300 series	PASS
9	Records per series	1,913 for every series	1,913 to 1,913	PASS
10	Series per date	300 on every date	300 to 300	PASS
11	Complete panel row count	573,900	573,900	PASS
12	Daily calendar continuity	0 non-daily gaps	0 non-daily gaps	PASS
13	Target data type	Numeric	int64	PASS
14	Target completeness	0 missing values	0 missing values	PASS
15	Target non-negativity	0 negative values	0 negative values	PASS
16	Input missingness pattern	Limited to sparse event-description fields	event_name_1, event_name_2, event_type_1, even...	PASS

✅ Input validation passed: 17 of 17 controls passed.

✅ Input Validation Result

The daily analytical base produced by Notebook 02 passed all input-contract and structural-readiness checks executed in Notebook 03:

- the dataset contains **573,900 daily observations** and **56 source columns**;

- the approved development scope contains exactly **300 SKU-store series** across **1,913 consecutive dates** from **29 January 2011 to 24 April 2016**;
- every series contains 1,913 daily records and every date contains all 300 series;
- the `sku_id` – `date` analytical key is complete and unique;
- weekly feature-engineering source fields are available with valid date and target data types;
- `units_sold` contains no missing or negative values;
- missingness is limited to event-description fields, where null values correctly represent non-event dates and are supported by a complete binary event indicator.

The daily analytical base is therefore approved for weekly aggregation and leakage-controlled feature engineering.  **Section 3 is complete.**



4. Weekly Modelling Base

This section aggregates the validated daily analytical base into one record per SKU-store retail week using the existing `wm_yr_wk` identifier. Weekly units sold will become the forecasting target, while commercial and calendar variables will be summarized according to their meaning.

Only retail weeks containing all seven daily observations will be retained.

4.1 Construct the Weekly Modelling Base and Target

Daily observations will be aggregated by SKU-store series and retail week. Weekly units sold will be calculated as the forecasting target, relevant price, promotion, event and SNAP information will be summarized, and stable product-store attributes will be retained.

SNAP modelling rule: The source `snap_flag` identifies SNAP activity in any tracked state. Because the modelling grain is SKU-store-week, the applicable SNAP indicator will instead be derived from each series's `state_id` and corresponding state-level SNAP field.

✅ Weekly modelling base constructed successfully.

metric											value
0	Candidate SKU-store-week records										82,200
1	Complete weekly records retained										81,900
2	Partial weekly records excluded										300
3	Complete retail weeks										273
4	SKU-store series retained										300
5	First complete week start										2011-01-29
6	Last complete week end										2016-04-22

	sku_id	wm_yr_wk	week_start_date	week_end_date	days_observed	weekly_units_sold	unit_price	promo_week_flag	event_days	snap_active_days	category	store_id
0	FOODS_1_0 01_CA_1	11101	2011-01-29	2011-02-04	7	10	2.00	0	0	4	FOODS	CA_1
1	FOODS_1_0 01_CA_1	11102	2011-02-05	2011-02-11	7	6	2.00	0	1	6	FOODS	CA_1
2	FOODS_1_0 01_CA_1	11103	2011-02-12	2011-02-18	7	10	2.00	0	1	0	FOODS	CA_1
3	FOODS_1_0 01_CA_1	11104	2011-02-19	2011-02-25	7	13	2.00	0	1	0	FOODS	CA_1
4	FOODS_1_0 01_CA_1	11105	2011-02-26	2011-03-04	7	15	2.00	0	0	4	FOODS	CA_1
5	FOODS_1_0 01_CA_1	11106	2011-03-05	2011-03-11	7	11	2.00	0	1	6	FOODS	CA_1
6	FOODS_1_0 01_CA_1	11107	2011-03-12	2011-03-18	7	7	2.00	0	2	0	FOODS	CA_1
7	FOODS_1_0 01_CA_1	11108	2011-03-19	2011-03-25	7	6	2.00	0	1	0	FOODS	CA_1
8	FOODS_1_0 01_CA_1	11109	2011-03-26	2011-04-01	7	5	2.00	0	0	1	FOODS	CA_1
9	FOODS_1_0 01_CA_1	11110	2011-04-02	2011-04-08	7	6	2.00	0	0	7	FOODS	CA_1

4.2 Verify the Weekly Grain

Focused checks will confirm the unique `sku_id - wm_yr_wk` grain, complete seven-day weeks, preservation of all 300 series and reconciliation of weekly target values with the corresponding retained daily demand.

	control	expected	observed	status
0	Unique SKU-store-week key	0 duplicates	0 duplicates	PASS
1	Complete seven-day weeks	0 invalid records	0 invalid records	PASS
2	Approved series retained	300 series	300 series	PASS
3	Records per series	273 for every series	273 to 273	PASS
4	Series per retail week	300 for every week	300 to 300	PASS
5	Continuous weekly chronology	0 weekly gaps	0 weekly gaps	PASS
6	Weekly target reconciliation	0 mismatches	0 mismatches	PASS

✅ Weekly modelling base validated: 7 of 7 controls passed.

✅ Weekly Modelling Base Result

The validated daily analytical base was successfully converted into **81,900 complete SKU-store-week observations** across **300 series and 273 consecutive retail weeks**. All records have a unique weekly key, complete seven-day coverage and continuous chronology. Weekly demand reconciles exactly with the corresponding daily demand, while 300 partial-week records were correctly excluded.

The weekly modelling base is approved for leakage-controlled feature engineering.

✅ **Section 4 is complete.**

5. Forecasting Feature Engineering

This section engineers the model-ready predictors required for weekly demand forecasting. The features will represent calendar position, historical demand, commercial history, known calendar drivers, point-in-time demand behaviour and product-store hierarchy context.

All demand-, price- and promotion-derived features will use information from weeks preceding the target week. Target-week calendar, scheduled-event and applicable SNAP information may be used only because it is known before the forecast is created.

5.1 Calendar and Seasonal Features

Calendar features will represent chronological progression and recurring annual seasonality. These predictors are known in advance and do not depend on future demand outcomes.

	wm_yr_wk	week_start_date	time_index	calendar_month	week_of_year	week_of_year_sin	week_of_year_cos
0	11101	2011-01-29	0	1	4	0.35	0.94
1	11102	2011-02-05	1	2	5	0.46	0.89
2	11103	2011-02-12	2	2	6	0.57	0.82
3	11104	2011-02-19	3	2	7	0.66	0.75
4	11105	2011-02-26	4	2	8	0.75	0.67
5	11106	2011-03-05	5	3	9	0.82	0.57
6	11107	2011-03-12	6	3	10	0.88	0.47
7	11108	2011-03-19	7	3	11	0.93	0.36
8	11109	2011-03-26	8	3	12	0.97	0.24
9	11110	2011-04-02	9	4	13	0.99	0.13
10	11111	2011-04-09	10	4	14	1.00	0.01
11	11112	2011-04-16	11	4	15	0.99	-0.11

Calendar Feature Summary

Five calendar predictors were created successfully. The `time_index` advances sequentially across retail weeks, while calendar month and ISO week identify seasonal position. The sine and cosine features represent annual seasonality continuously across the year-end boundary.

Because all five features are derived solely from the retail-week start date, they are known before forecasting and introduce no target leakage.

5.2 Historical Demand Features

Demand lags and shifted historical statistics will summarize recent, medium-term and annual demand patterns. Each calculation will exclude the target week's demand so that only preceding observations contribute to its predictors.

Multi-step forecasting rule: During an eight-week validation or future forecast, demand-derived features will be updated recursively using preceding observations or model predictions. Actual demand from inside the forecast period must never enter subsequent predictors.

	sku_id	wm_yr_wk	week_start_date	weekly_units_sold	demand_lag_1	demand_lag_4	demand_lag_8	demand_lag_13	demand_lag_52	demand_rolling_mean_4	demand_rolling_mean_13	demand_rolling_mean_52	demand_rolling_std_13
52	FOODS_1_001_CA_1	11201	2012-01-28	7	2.00	6.00	12.00	8.00	10.00	4.75	8.69	6.27	6.06
53	FOODS_1_001_CA_1	11202	2012-02-04	5	7.00	4.00	6.00	7.00	6.00	5.00	8.62	6.21	6.08
54	FOODS_1_001_CA_1	11203	2012-02-11	6	5.00	7.00	26.00	8.00	10.00	5.25	8.46	6.19	6.15
55	FOODS_1_001_CA_1	11204	2012-02-18	6	6.00	2.00	4.00	10.00	13.00	5.00	8.31	6.12	6.18
56	FOODS_1_001_CA_1	11205	2012-02-25	8	6.00	7.00	6.00	13.00	15.00	6.00	8.00	5.98	6.19
57	FOODS_1_001_CA_1	11206	2012-03-03	6	8.00	5.00	4.00	12.00	11.00	6.25	7.62	5.85	6.01
58	FOODS_1_001_CA_1	11207	2012-03-10	8	6.00	6.00	7.00	6.00	7.00	6.50	7.15	5.75	5.87
59	FOODS_1_001_CA_1	11208	2012-03-17	5	8.00	6.00	2.00	26.00	6.00	7.00	7.31	5.77	5.86

Historical Demand Feature Summary

Fifteen historical-demand predictors were created: seven demand lags, five shifted rolling means and three shifted rolling standard deviations. The preview begins at the first observation with 52 weeks of preceding history.

All rolling calculations exclude the target week through a one-week shift. Earlier missing values are therefore expected structural missingness and will be handled by the history-eligibility rule in Section 6.

5.3 Commercial and Known Calendar Drivers

Historical price and promotion features will use information from weeks preceding the target week. Scheduled-event and state-applicable SNAP features may describe the target week because their timing is known in advance.

Forecast-horizon rule: Future price and promotion outcomes are not assumed to be known. Their historical features are therefore shifted by eight weeks, keeping them outside the complete forecast block. Target-week event and applicable SNAP information may be used directly because it is known in advance.

	sku_id	wm_yr_wk	week_start_date	unit_price	price_lag_8	price_change_lag_8	price_rolling_mean_4_lag_8	promo_week_flag	promo_lag_8	promo_rolling_rate_4_lag_8	event_days	snap_active_days
52	FOODS_1_001_CA_1	11201	2012-01-28	2.00	2.00	0.00	2.00	0	0.00	0.00	0	3
53	FOODS_1_001_CA_1	11202	2012-02-04	2.00	2.00	0.00	2.00	0	0.00	0.00	1	7
54	FOODS_1_001_CA_1	11203	2012-02-11	2.00	2.00	0.00	2.00	0	0.00	0.00	1	0
55	FOODS_1_001_CA_1	11204	2012-02-18	2.00	2.00	0.00	2.00	0	0.00	0.00	2	0
56	FOODS_1_001_CA_1	11205	2012-02-25	2.00	2.00	0.00	2.00	0	0.00	0.00	1	2
57	FOODS_1_001_CA_1	11206	2012-03-03	2.00	2.00	0.00	2.00	0	0.00	0.00	1	7
58	FOODS_1_001_CA_1	11207	2012-03-10	2.00	2.00	0.00	2.00	0	0.00	0.00	0	1
59	FOODS_1_001_CA_1	11208	2012-03-17	2.00	2.00	0.00	2.00	0	0.00	0.00	1	0

Commercial and Calendar Driver Summary

Five historical commercial predictors and six known calendar drivers were prepared. Unavailable zero-price records are not treated as genuine selling prices; price features instead use the last positive price observed by the historical cutoff. Price change is recorded as zero when no valid prior comparison exists, while the lagged price remains zero when no positive price has yet been observed.

All price and promotion features remain at least eight weeks behind the target. Scheduled-event and applicable SNAP features describe the target week directly because their timing is known before forecasting.

5.4 Demand-Behaviour and Hierarchy Features

Point-in-time demand-behaviour features will describe prior selling activity, demand recency and intermittency without using full-series classifications. Stable product and store hierarchy fields will be retained as categorical context for the global forecasting models.

	sku_id	wm_yr_wk	week_start_date	weekly_units_sold	prior_active_week_count	weeks_since_last_sale	active_week_rate_13	item_id	category	department	store_id	state_id
52	FOODS_1_01_CA_1	11201	2012-01-28	7	40	1	1.00	FOODS_1_01	FOODS	FOODS_1	CA_1	CA
53	FOODS_1_01_CA_1	11202	2012-02-04	5	41	1	1.00	FOODS_1_01	FOODS	FOODS_1	CA_1	CA
54	FOODS_1_01_CA_1	11203	2012-02-11	6	42	1	1.00	FOODS_1_01	FOODS	FOODS_1	CA_1	CA
55	FOODS_1_01_CA_1	11204	2012-02-18	6	43	1	1.00	FOODS_1_01	FOODS	FOODS_1	CA_1	CA
56	FOODS_1_01_CA_1	11205	2012-02-25	8	44	1	1.00	FOODS_1_01	FOODS	FOODS_1	CA_1	CA
57	FOODS_1_01_CA_1	11206	2012-03-03	6	45	1	1.00	FOODS_1_01	FOODS	FOODS_1	CA_1	CA
58	FOODS_1_01_CA_1	11207	2012-03-10	8	46	1	1.00	FOODS_1_01	FOODS	FOODS_1	CA_1	CA
59	FOODS_1_01_CA_1	11208	2012-03-17	5	47	1	1.00	FOODS_1_01	FOODS	FOODS_1	CA_1	CA



Demand-Behaviour and Hierarchy Feature Summary

Three point-in-time demand-behaviour predictors were created to represent prior selling activity, demand recency and recent intermittency. For the displayed observations, continuous recent sales cause the prior active-week count to increase, the weeks since the last sale to remain at one and the preceding 13-week activity rate to remain at 1.00.

Five stable product-store hierarchy fields were retained for encoding within the later modelling pipeline. All demand-behaviour calculations exclude the target week's outcome and therefore introduce no target leakage.

✔ **Section 5 is complete.**

6. Final Modelling Dataset

This section applies the minimum-history rule and organizes eligible weekly observations into the identifiers, target and approved numerical and categorical predictors required for modelling.

Missing values produced before sufficient lag or rolling history is available are structural and will not be replaced with zero.

6.1 Determine History Eligibility

The number of complete retail weeks preceding each observation will be calculated. Observations with at least 52 preceding weeks will be retained for the primary forecasting workflow, ensuring that every approved demand-history feature is available.

				metric	value
		0	Complete weekly records assessed		81,900
		1	Structural-history records excluded		15,600
		2	History-eligible records retained		66,300
		3	History-eligible retail weeks		221
		4	SKU-store series retained		300
		5	Eligible records per series		221 to 221
		6	First eligible week start		2012-01-28
		7	Last eligible week end		2016-04-22

	sku_id	wm_yr_wk	week_start_date	history_weeks_available	history_eligible	demand_lag_52	demand_rolling_mean_52
50	FOODS_1_001_CA_1	11151	2012-01-14	50	False	NaN	NaN
51	FOODS_1_001_CA_1	11152	2012-01-21	51	False	NaN	NaN
52	FOODS_1_001_CA_1	11201	2012-01-28	52	True	10.00	6.27
53	FOODS_1_001_CA_1	11202	2012-02-04	53	True	6.00	6.21
54	FOODS_1_001_CA_1	11203	2012-02-11	54	True	10.00	6.19

History Eligibility Summary

The 52-week minimum-history rule excluded 15,600 structural-history observations and retained **66,300 eligible records** across all **300 SKU-store series**. Each series contributes 221 eligible weeks spanning 28 January 2012 through 22 April 2016.

The eligibility boundary behaves correctly: observations with 50 or 51 preceding weeks remain ineligible, while the observation with 52 preceding weeks is the first to contain complete annual-history features.

6.2 Finalize the Feature Schema

Eligible observations will be checked for remaining missing or non-finite predictor values. Identifiers and the weekly target will remain separate from the approved numerical and categorical feature lists used by the modelling pipeline.

	metric	value
0	History-eligible records	66,300
1	Identifier columns	4
2	Target columns	1
3	Numerical predictors	34
4	Categorical predictors	5
5	Total predictors	39
6	Final modelling columns	44
7	Missing predictor values	0
8	Non-finite numerical values	0

✅ Final modelling schema constructed successfully.

Final Modelling Schema Summary

The final in-memory modelling dataset contains **66,300 history-eligible weekly observations** and **44 columns**: four identifiers, one target, 34 numerical predictors and five categorical predictors.

Schema checks confirmed that all required columns are present, identifiers and the target are excluded from the predictor set, and no duplicated schema columns exist. The 39 predictors contain no missing or non-finite values.

The dataset has not yet been exported; complete feature and leakage validation will be performed in Section 7 before export.

✅ **Section 6 is complete.**

✅ 7. Feature Validation and Leakage Audit

This section validates the completed modelling dataset. Focused checks will verify structural integrity, feature-calculation accuracy and the absence of prohibited future information.

7.1 Structural Validation

The final dataset will be checked for unique SKU-week keys, chronological ordering within each series, complete retention of eligible series, valid non-negative targets, distinct feature names, complete identifiers and finite predictor values.

	check	result
0	Duplicated SKU-week keys	0
1	Chronological ordering within series	Valid
2	Eligible series retained	300 of 300
3	Records outside the eligible set	0
4	Missing identifier values	0
5	Missing target values	0
6	Negative target values	0
7	Duplicated feature names	0
8	Missing predictor values	0
9	Non-finite numerical values	0

✅ Structural validation passed successfully.

Structural Validation Summary

The final modelling dataset contains unique SKU-week observations in strict chronological order and retains all **300 history-eligible SKU-store series**.

All records belong to the eligible observation set. Identifiers, targets and predictors are complete; targets are non-negative; feature names are distinct; and all numerical values are finite.

7.2 Feature Calculation Checks

Selected observations will be reconciled with their preceding historical source values to verify the alignment of demand lags, shifted rolling statistics, SNAP indicators and demand-recency features.

		check	records_checked	mismatched_records	maximum_absolute_difference
0		One-week demand lag	66,300	0	0.000000
1		Annual demand lag	66,300	0	0.000000
2		Shifted 4-week demand mean	66,300	0	0.000000
3		Shifted 13-week demand deviation	66,300	0	0.000004
4		Applicable SNAP-active days	66,300	0	0.000000
5		Weeks since preceding sale	66,300	0	0.000000

	sku_id	wm_yr_wk	week_start_date	feature	stored_value	source_value	matches
0	FOODS_1_001_CA_1	11201	2012-01-28	demand_lag_1	2.00	2.00	True
1	FOODS_1_001_CA_1	11201	2012-01-28	demand_lag_52	10.00	10.00	True
2	FOODS_1_001_CA_1	11201	2012-01-28	demand_rolling_mean_4	4.75	4.75	True
3	FOODS_1_001_CA_1	11201	2012-01-28	demand_rolling_std_13	6.06	6.06	True
4	FOODS_1_001_CA_1	11201	2012-01-28	snap_active_days	3.00	3.00	True
5	FOODS_1_001_CA_1	11201	2012-01-28	weeks_since_last_sale	1.00	1.00	True
6	FOODS_1_001_CA_1	11406	2014-03-08	demand_lag_1	10.00	10.00	True
7	FOODS_1_001_CA_1	11406	2014-03-08	demand_lag_52	5.00	5.00	True
8	FOODS_1_001_CA_1	11406	2014-03-08	demand_rolling_mean_4	6.50	6.50	True
9	FOODS_1_001_CA_1	11406	2014-03-08	demand_rolling_std_13	3.39	3.39	True
10	FOODS_1_001_CA_1	11406	2014-03-08	snap_active_days	3.00	3.00	True
11	FOODS_1_001_CA_1	11406	2014-03-08	weeks_since_last_sale	1.00	1.00	True
12	FOODS_1_001_CA_1	11612	2016-04-16	demand_lag_1	6.00	6.00	True
13	FOODS_1_001_CA_1	11612	2016-04-16	demand_lag_52	2.00	2.00	True
14	FOODS_1_001_CA_1	11612	2016-04-16	demand_rolling_mean_4	8.25	8.25	True
15	FOODS_1_001_CA_1	11612	2016-04-16	demand_rolling_std_13	2.61	2.61	True
16	FOODS_1_001_CA_1	11612	2016-04-16	snap_active_days	0.00	0.00	True
17	FOODS_1_001_CA_1	11612	2016-04-16	weeks_since_last_sale	1.00	1.00	True

✔ Feature calculation checks passed successfully.

Feature Calculation Check Summary

Six representative features were independently recalculated and reconciled across all **66,300 modelling observations**. The checks covered short- and annual-demand lags, shifted rolling statistics, applicable SNAP activity and demand recency.

No mismatches were found, and every maximum absolute difference was zero. Representative observations from the beginning, middle and end of an example series also matched their historical source values exactly.

7.3 Leakage Audit

The approved feature list and critical feature-generation rules will be audited to confirm that each predictor uses only preceding information or target-week information known independently in advance. Prohibited fields and unsafe alignments will be rejected through assertions.

	feature_group	availability_rule	feature_count
0	Calendar features	Target-week timing known in advance	5
1	Historical demand features	Preceding demand observations only	15
2	Historical commercial features	At least eight weeks behind the target	5
3	Known calendar drivers	Target-week schedules known in advance	6
4	Demand-behaviour features	Preceding demand activity only	3
5	Static hierarchy features	Static product and store metadata	5

	feature	records_checked	mismatched_records
0	price_lag_8	66,300	0
1	price_change_lag_8	66,300	0
2	price_rolling_mean_4_lag_8	66,300	0
3	promo_lag_8	66,300	0
4	promo_rolling_rate_4_lag_8	66,300	0

	check	result
0	Predictors audited	39
1	Availability-registry coverage	39 of 39
2	Feature-group overlaps	0
3	Unregistered predictors	0
4	Unexpected registered features	0
5	Prohibited predictors	0
6	Unsafe future or lead names	0
7	Minimum demand lag	1 week
8	Commercial-history shift	8 weeks
9	Commercial-history mismatches	0
10	Approved known calendar drivers	6 of 6
11	Previous calculation mismatches	0
12	Multi-step demand requirement	Recursive inference required

✅ Leakage audit passed successfully.

Leakage Audit Summary

All **39 predictors** were assigned exactly once to one of six approved availability groups. No unregistered, overlapping, prohibited or future-named predictors were found.

Historical demand features use only preceding outcomes, while all five commercial-history features were independently reconciled across **66,300 observations** with zero mismatches and remain at least eight weeks behind the target. All six target-week calendar drivers are approved information known independently in advance.

All demand-derived predictors—including demand lags, rolling-demand statistics, prior active-week counts, demand recency and active-week rates—are safe for multi-step forecasting only when future notebooks recompute them recursively from information available at the forecast origin and from prior model predictions.

Precomputed feature values from inside a validation, test or future forecast block must never be used to predict its later horizons because those values may contain actual intermediate demand that was not available at the forecast origin.

✔ Section 7 is complete. The validated modelling dataset is ready for controlled export.

8. Export and Validate Modelling Outputs

This section exports the validated weekly modelling dataset and its supporting feature metadata. The persisted outputs will then be reloaded and verified as a reproducible handoff to the forecasting stage.

8.1 Export Modelling Outputs

The final modelling dataset, complete predictor list, categorical predictor list and feature dictionary will be exported to the dedicated `data/modeling` directory. The directory will be created only when the validated outputs are ready to be saved.

✔ Export metadata prepared successfully.

Dataset columns documented : 44

Model predictors documented: 39

	output	file_name	content	file_size_kb	status
0	Weekly modelling dataset	weekly_features_final.csv	66,300 rows x 44 columns	16,833.70	Exported
1	Model feature list	model_feature_list.json	39 features	0.88	Exported
2	Categorical feature list	categorical_features.json	5 features	0.08	Exported
3	Feature dictionary	feature_dictionary.csv	44 rows x 8 columns	4.91	Exported

✔ Modelling outputs exported successfully.

Output directory: C:\Users\hp\Project FORESIGHT\data\modeling

Modelling Output Export Summary

The validated modelling dataset was exported with **66,300 observations and 44 columns**, including **39 approved predictors** and **5 categorical predictors**.

The complete predictor list, categorical predictor list and **44-row feature dictionary** were also saved successfully. All four files are non-empty and available in the dedicated `data/modeling` directory for the forecasting-stage handoff.

8.2 Reload and Validate Outputs

All exported files will be reloaded to confirm their accessibility, dimensions, schemas, unique keys and consistency with the approved feature definitions. The final validation will also confirm that prohibited target-derived or future-information fields are absent from the model predictors.

	output	file_name	reloaded_content	status
0	Weekly modelling dataset	weekly_features_final.csv	66,300 rows x 44 columns	Reloaded
1	Model feature list	model_feature_list.json	39 features	Reloaded
2	Categorical feature list	categorical_features.json	5 features	Reloaded
3	Feature dictionary	feature_dictionary.csv	44 rows x 8 columns	Reloaded

✔ All modelling outputs reloaded successfully.

	check	result
0	Dataset dimensions	66,300 rows x 44 columns
1	Column order	Valid
2	Round-trip values	Match
3	Duplicated SKU-week keys	0
4	Chronological ordering within series	Valid
5	SKU-store series retained	300 of 300
6	Missing identifier values	0
7	Missing target values	0
8	Missing predictor values	0
9	Negative target values	0
10	Non-finite numerical values	0

✅ Reloaded dataset integrity validated successfully.

	check	result
0	Model feature-list agreement	39 of 39
1	Categorical feature-list agreement	5 of 5
2	Feature-dictionary rows	44 of 44
3	Feature-dictionary schema	8 of 8
4	Dataset-column dictionary coverage	44 of 44
5	Identifier-role alignment	Valid
6	Target-role alignment	Valid
7	Predictor-role alignment	Valid
8	Model-feature flag alignment	Valid
9	Categorical-feature flag alignment	Valid
10	Missing dictionary values	0
11	Non-numeric numerical predictors	0
12	Prohibited predictors	0
13	Unsafe future or lead names	0
14	Forecasting handoff	Ready

✅ Reloaded modelling outputs validated successfully.

✅ Notebook 03 handoff is ready for forecasting.

	Validation Area	Metric	Result	Requirement	Status
0	Dataset Scope	Eligible modelling observations	66,300	66,300	PASS
1	Dataset Scope	Dataset columns	44	44	PASS
2	Dataset Scope	SKU-store series	300	300	PASS
3	Feature Registration	Approved model predictors	39	39	PASS
4	Feature Registration	Categorical predictors	5	5	PASS
5	Feature Registration	Feature-availability groups	6	6	PASS
6	Feature Metadata	Feature-dictionary rows	44	44	PASS
7	Dataset Integrity	Column order	Valid	Valid	PASS
8	Dataset Integrity	Duplicated SKU-week keys	0	0	PASS
9	Dataset Integrity	Chronological ordering	Valid	Valid	PASS
10	Dataset Integrity	Missing identifier values	0	0	PASS
11	Dataset Integrity	Missing target values	0	0	PASS
12	Dataset Integrity	Missing predictor values	0	0	PASS
13	Dataset Integrity	Negative target values	0	0	PASS
14	Dataset Integrity	Non-finite numerical values	0	0	PASS
15	Persistence	CSV round-trip values	Match	Match	PASS
16	Feature Metadata	Missing dictionary values	0	0	PASS
17	Feature Metadata	Non-numeric numerical predictors	0	0	PASS
18	Leakage Control	Prohibited predictors	0	0	PASS
19	Leakage Control	Unsafe future or lead names	0	0	PASS
20	Final Handoff	Forecasting handoff	Ready	Ready	PASS

✅ Feature-validation summary exported successfully.

Validation checks: 21

Checks passed : 21

Output file : C:\Users\hp\Project FORESIGHT\reports\tables\feature_validation_summary.csv

Reload and Validation Summary

All four modelling outputs were reloaded successfully and matched their approved definitions.

The reloaded dataset retained all **66,300 observations, 44 columns and 300 SKU-store series**. Column order and round-trip values matched the original dataset, chronological ordering remained valid and no duplicated SKU-week keys were found.

All identifiers, target values and predictors were complete. No negative demand values, non-finite numerical values, prohibited predictors or unsafe future-information fields were detected.

The exported metadata also remained fully aligned, including all **39 model predictors, 5 categorical predictors** and the complete **44-row feature dictionary**.

✔ **Section 8 is complete. Notebook 03 has produced a validated and reproducible modelling-data handoff ready for forecasting.**

9. Conclusion and Notebook 04 Handoff

Notebook 03 transformed the validated daily analytical data into a weekly, leakage-safe modelling dataset for Project FORESIGHT. The notebook completed weekly aggregation, target construction, feature engineering, history-eligibility filtering, predictor registration, leakage auditing and final output validation.

9.1 Work Completed and Key Outcomes

Work Completed

Notebook 03 successfully:

- converted the validated daily analytical base into a consistent SKU-store-week modelling structure;
- created the weekly demand target;
- engineered calendar, cyclical, historical-demand, rolling, expanding, trend, momentum, commercial-history, demand-behaviour and hierarchy features;
- applied history-eligibility requirements before selecting modelling observations;
- documented every identifier, target and predictor in a complete feature dictionary;
- audited feature availability and historical alignment;
- exported and reloaded all modelling outputs;
- validated dimensions, ordering, keys, schemas, completeness and round-trip values.

Final Modelling Dataset

The completed modelling dataset contains:

Measure	Final Result
Eligible modelling observations	66,300
Dataset columns	44
SKU-store series	300
Approved model predictors	39
Categorical predictors	5
Feature-availability groups	6

All 300 SKU-store series were retained. The dataset contains no duplicated SKU-week keys, missing identifiers, missing target values, missing predictors, negative target values or non-finite numerical values.

9.2 Leakage Controls and Final Outputs

Leakage-Control Results

All **39 predictors** were assigned exactly once to an approved availability group. No overlapping, unregistered, prohibited or unsafe future-named predictors were detected.

Historical-demand predictors use only preceding outcomes. All five commercial-history features were independently reconciled across **66,300 observations** with zero mismatches and remain at least eight weeks behind the target period. All six target-week calendar drivers are approved information known independently in advance.

All demand-derived predictors—including demand lags, rolling-demand statistics, prior active-week counts, demand recency and active-week rates—remain valid for multi-step forecasting only when future notebooks recompute them recursively using information available at the forecast origin and prior model predictions.

Precomputed feature values from inside a validation, test or future forecast block must never be used to predict its later horizons because those values may contain actual intermediate demand unavailable at the forecast origin.

Validated Outputs

The following outputs were exported to `data/modeling` and successfully reloaded:

Output File	Purpose
<code>weekly_features_final.csv</code>	Final weekly modelling dataset
<code>model_feature_list.json</code>	Ordered list of 39 approved predictors
<code>categorical_features.json</code>	Ordered list of 5 categorical predictors
<code>feature_dictionary.csv</code>	Metadata for all 44 dataset columns

The exported files matched their approved dimensions, schemas, feature definitions and original in-memory values.

Scope Boundary

Notebook 03 prepares the modelling data but does not train, compare or select forecasting models. Chronological evaluation, baseline forecasting and performance measurement belong to Notebook 04.

9.3 Notebook 04 Handoff

Project FORESIGHT will now proceed to:

`04_Baseline_Forecasting.ipynb`

Notebook 04 will:

- reload and verify the validated modelling outputs;
- formalize the eight-week forecast horizon;
- establish chronological training, validation and test periods;
- define and apply WAPE and Forecast Bias as the locked primary evaluation metrics;
- implement naive, seasonal-naive, moving-average and exponential-smoothing baselines;
- perform rolling-origin validation;
- compare baseline performance and forecasting behaviour;
- select the champion baseline for later machine-learning comparison;
- export baseline forecasts, metrics and configuration metadata.

The chronological design must avoid random splitting and must preserve the leakage controls established in Notebook 03. At every rolling origin, all demand-derived predictors for later horizons must be rebuilt recursively using information available at that origin and prior model predictions. Actual intermediate demand from the validation, test or future forecast block must never be used.

WAPE and Forecast Bias must provide the primary evaluation view, while the seasonal-naive model must remain the principal benchmark for subsequent model evaluation.

Final Status: Notebook 03 — Feature Engineering completed successfully.

✓ **Section 9 is complete.**

✓ **Notebook 03 has produced a validated, reproducible and leakage-safe modelling-data handoff ready for baseline forecasting.**
